

■ Document your SQL queries and procedures to facilitate maintenance and collaboration.

In this chapter, we have introduced SQL, the standard language for managing and manipulating relational databases. You have learned how to set up your SQL environment, explored basic

syntax, and written various SQL queries to manage data. We also discussed industry trends and best practices for SQL development, ensuring you are well-equipped to build efficient and secure database applications. As you continue your coding journey, SQL will be an invaluable tool for data management and analysis. 

SQL COMMANDS AND FUNCTIONS CHEAT SHEET		
Here's a cheat sheet that summarizes essential SQL commands and functions. This table includes descriptions and syntax examples to help you quickly reference and utilize SQL commands in your projects.		
Command/Function	Description	Syntax Example
CREATE DATABASE	Creates a new database.	CREATE DATABASE my_database;
DROP DATABASE	Deletes an existing database.	DROP DATABASE my_database;
CREATE TABLE	Creates a new table.	CREATE TABLE employees (id INT, name VARCHAR(50));
DROP TABLE	Deletes an existing table.	DROP TABLE employees;
ALTER TABLE	Modifies an existing table structure.	ALTER TABLE employees ADD COLUMN email VARCHAR(100);
INSERT INTO	Inserts new data into a table.	INSERT INTO employees (id, name) VALUES (1, 'John Doe');
SELECT	Retrieves data from a database.	SELECT * FROM employees;
WHERE	Filters records.	SELECT * FROM employees WHERE id = 1;
UPDATE	Updates existing data in a table.	UPDATE employees SET name = 'Jane Doe' WHERE id = 1;
DELETE	Deletes existing data from a table.	DELETE FROM employees WHERE id = 1;
JOIN	Combines rows from two or more tables.	SELECT * FROM employees JOIN departments ON employees.dept_id = departments.id;
ORDER BY	Sorts the result set.	SELECT * FROM employees ORDER BY name ASC;
GROUP BY	Groups rows that have the same values.	SELECT COUNT(*), department FROM employees GROUP BY department;
HAVING	Filters groups based on a condition.	SELECT COUNT(*), department FROM employees GROUP BY department HAVING COUNT(*) > 1;
LIMIT	Limits the number of rows returned.	SELECT * FROM employees LIMIT 10;
DISTINCT	Selects unique values.	SELECT DISTINCT department FROM employees;
COUNT	Returns the number of rows that match a condition.	SELECT COUNT(*) FROM employees;
AVG	Returns the average value of a numeric column.	SELECT AVG(salary) FROM employees;
SUM	Returns the total sum of a numeric column.	SELECT SUM(salary) FROM employees;
MIN	Returns the smallest value of the selected column.	SELECT MIN(salary) FROM employees;
MAX	Returns the largest value of the selected column.	SELECT MAX(salary) FROM employees;